

Will AI Replace Drivers and Pilots?

The planning guide — the full scores by track, the six factors behind them, the honest trade-offs, and what to ask before you commit.



4.2

AI EXPOSURE · AIRLINE PILOT

where 10 is most at risk

For anyone still choosing — read it straight through.



Before you read this

You're weighing a path into transport and logistics — driving, specialist vehicle work, the technician route, or the planning side — and you want to know what that choice actually looks like in an AI economy. That's the whole job of this guide.

It's written to you directly, whether you are the one choosing or a parent thinking it through alongside them. Where it says "you," a parent can read "them"; it's the same decision from either side.

What this is. A facts document, not a recommendation. It sets out what's known, where the evidence is contested, and where we might be wrong — and then how to choose well given all of it.

If you're the parent, a note on how to use it. Don't lead with the scores. Open a conversation with a risk number and one of two things happens: they shut down, or they get defensive about a choice they're already leaning toward. Read it yourself first, then share it as *information you both react to*, not a verdict you've reached. They're choosing, not you — the job is to widen what's been considered, not to narrow it.

And one thing specific to transport. This guide contradicts the single most confidently repeated prediction of the last decade — that driving jobs would be the first to go. Nine years after that prediction became mainstream, the data says something different, and the office is automating considerably faster than the cab. If the instinct is that driving is a dead-end career because of self-driving trucks, the section on the prediction that didn't happen is the one to read. The honest problems here are real, and they are about working conditions rather than about robots.

The short answer

The prediction that driving jobs would go first has not happened — and the planning jobs behind them are automating instead.

Emergency and specialist vehicle operation scores 2.8 out of 10. Long-haul trucking scores 4.5. Logistics planning and scheduling scores 7.3 — more exposed than any driving role we measure.

The person behind the wheel is better protected than the person planning their route. That is the opposite of what almost everyone expected, and the sections that follow explain why — which makes which version of transport you aim at most of the decision in front of you.

The scores

TRACK	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Emergency & specialist vehicle operation	2.4	2.6	2.8	+0.4	Very low
Local delivery & multi-stop	3.0	3.2	3.4	+0.4	Low
Airline pilot	3.1	3.4	3.6	+0.5	Low
Rail operation	3.6	3.9	4.2	+0.6	Low– Moderate
Long-haul trucking	3.8	4.2	4.5	+0.7	Moderate
Freight brokerage & dispatch	5.8	6.4	6.8	+1.0	Moderate– High
Logistics planning & scheduling	6.2	6.9	7.3	+1.1	High

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, a service electrician scores 2.5, bedside nursing 2.8, and a business analyst 7.6.

Read the table this way: the split runs between the vehicle and the desk, not between skilled and unskilled work. Every role involving a physical vehicle in an unpredictable environment sits below 4.5. Every role planning that movement from a screen sits above 6.8.

The prediction that didn't happen

This deserves the most space in the guide, because it is the most widely believed claim about any career in this index — and because the evidence against it is unusually specific.

The prediction: since roughly 2017, autonomous trucking has been described as imminent, and driving has been the standard example of the job AI would eliminate first.

Where it actually stands in mid-2026:

- There are still **no commercially scaled, fully driverless freight operations** on US highways.
- Aurora, Kodiak and Plus run limited driver-out pilots on specific corridors in Texas and Arizona.
- Those pilots account for **fewer than 50 trucks combined**, against approximately **4 million Class 8 trucks** operating nationally.
- Even optimistic industry forecasts put commercial Level 4 autonomy at **5–10% of long-haul lanes by 2032**.

And three structural limits matter more than the current numbers.

The transfer-hub model preserves human drivers at both ends. Autonomous operation targets the highway middle of a journey. **First mile and last mile — the parts involving loading docks, city streets and customer sites — remain human by design, not by delay.**

It only addresses one kind of freight. Industry analysis puts roughly **60% of trucking work** in categories autonomy does not touch: dock visits, multi-stop deliveries, hazmat, oversize loads, and anything requiring interaction at a customer site.

And it doesn't work everywhere. Current operations are concentrated in Sun Belt states. Winter conditions remain unsolved.

So the honest position is not that autonomy isn't coming — it is that after nine years it accounts for roughly **one truck in eighty thousand**, and the parts of the job it addresses least are the parts most drivers actually do.

Why we still score long-haul at 4.5 rather than lower. Because the trajectory is real even if the timeline was wrong, and because long-haul highway driving is the single most exposed driving category. **A student choosing driving work should weight local, specialist and multi-stop work more heavily than over-the-road long-haul — which is also where the pay growth has been strongest.**

The office is automating faster than the cab

Here is the finding that matters most, and almost nobody expects it.

While autonomous trucks were accumulating fifty vehicles, the back office was being rebuilt.

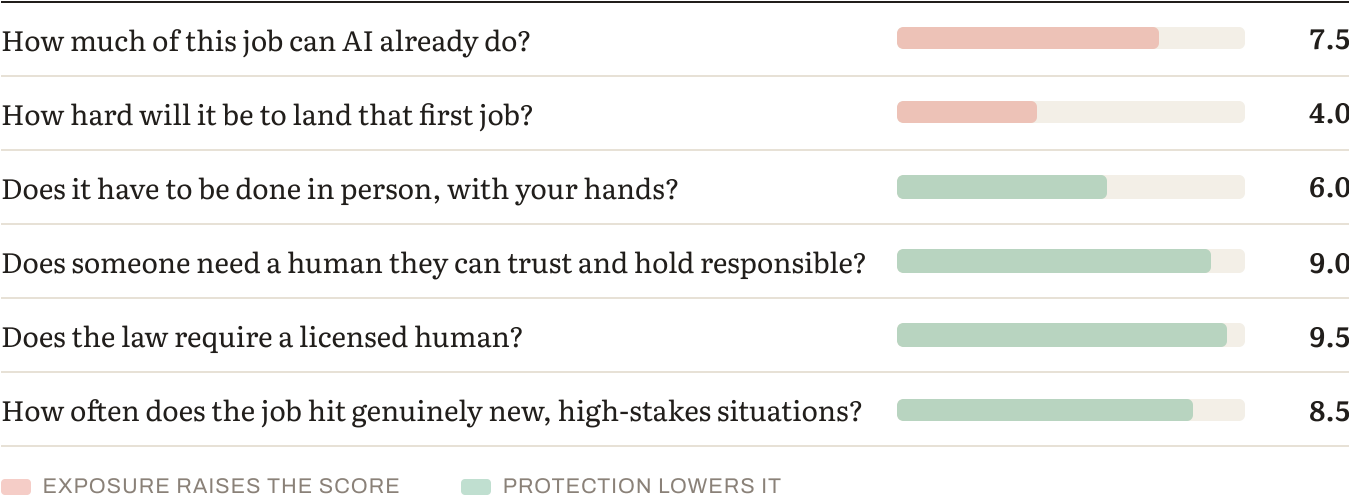
Warehouse management systems, transportation management systems and AI-driven demand planning are now standard at mid-size and enterprise logistics operations. Autonomous picking has moved out of pilots into deployment. And industry analysis notes that back-office roles including dispatchers, safety coordinators and logistics administrators can be partially offloaded, reducing overhead on those functions by 30–50%.

Route planning is the clearest case. Optimizing a multi-stop route against traffic, windows, weight limits and driver hours is a constrained optimization problem with structured inputs and a measurable correct answer. That is close to a definition of automatable work, and it used to be a career.

So the shape of transport employment is inverting. The job everyone predicted would disappear is holding. The jobs that coordinate it — dispatch, planning, brokerage, administration — are the exposed ones.

The general lesson, and it runs through this entire index: exposure follows the shape of the work, not its status. Physical work in unpredictable settings is protected regardless of how skilled it is considered. Screen work on structured inputs is exposed regardless of how professional it looks.

Why driving holds — the six factors



Here's how local delivery and multi-stop driving answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

Route optimization and the paperwork. Not the driving, and not the delivery.

Route planning and sequencing. Load matching and capacity planning. Hours-of-service compliance tracking. Proof-of-delivery processing. Fuel and maintenance scheduling.

What resists: reversing a trailer onto a difficult dock; deciding whether a road is safe in poor conditions; securing an unusual load; dealing with a customer whose site access isn't what the system says; and everything that happens when the plan meets reality.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Among the best on-ramps we measure. BLS projects roughly **237,600 annual openings** for heavy and tractor-trailer drivers through 2034. Training is short, licensed, and often employer-funded.

But an easy on-ramp hides a fork worth seeing now. The openings are easiest of all into long-haul over-the-road work — the segment with the worst turnover (90–95% at large carriers) and the highest exposure among driving roles (4.5). Taking the first seat that's hiring is not the same as entering the sustainable version of the career. The way through is to aim deliberately at local, LTL and specialist work — better protected (down to 2.8), better pay growth, and where people actually stay — rather than at whichever segment is recruiting hardest. Choosing the segment and the carrier well is exactly what the second half of this guide is about.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Entirely — and, crucially, in settings that vary constantly.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Moderately. Safety accountability is real and regulated; commercial trust matters most in specialist and customer-facing work.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

Substantially — and the section on pilots explains why this matters more in aviation than anywhere.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Frequently in local and specialist work; less so in repetitive line-haul. That difference is most of the spread between 2.8 and 4.5.

Pilots: protected by regulation, not by manual skill

Airline pilots deserve a note, because the reasoning is instructive and widely misunderstood.

Modern commercial aircraft can already fly, navigate and land largely automatically. The manual flying skill that people imagine protects pilots is not, in fact, what protects them.

What protects them is a regulatory and liability architecture built over a century, in which two qualified humans are legally required in the cockpit, certified by national authorities, subject to recurrent testing, and answerable for the aircraft.

That architecture moves at the speed of international treaty, national regulation and public confidence — and it is not going to be revised quickly regardless of technical capability.

And judgment under genuine novelty remains the pilot's core function: the failure the checklist doesn't cover, the weather that isn't in the forecast, the decision to divert.

The transferable lesson: when a technology can already perform a task and humans still do it, look at the legal architecture. Aviation is the clearest case in this index of regulation doing the protective work that people mistakenly attribute to skill.

What the trend shows

Local delivery: 3.0 → 3.2 → 3.4. Up 0.4 in three years — among the slowest in the index.

Logistics planning: 6.2 → 6.9 → 7.3. Up 1.1.

The gap widened from 3.2 points to 3.9. And the direction is the story: in three years of measurement, the desk-based half of this industry moved nearly three times as fast as the vehicle-based half.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Strong	The one to watch	Autonomy is real but slow, geographically limited, and doesn't address the 60% of work involving docks and customers.
Regulatory / licensing	Strong, especially aviation	Most durable	Aviation's architecture moves at treaty speed. Road transport licensing is meaningful but lighter.
Judgment under novelty	Strong in local and specialist work	Durable	Every site is different; the plan meets reality daily.
Human trust / accountability	Moderate	Durable where customer-facing	Safety accountability is regulated and personal.

The honest read. Driving's protection is genuine and it is not permanent — it is a timeline rather than a wall. Someone entering long-haul work today should expect the landscape to shift within their career. Someone entering specialist, local or emergency vehicle work is on considerably firmer ground.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Planning your own route	The route arrives optimized
Paperwork and logs	Automated compliance tracking
Finding backhaul loads	Matched automatically
Dispatching by phone	Largely automated — and this was the career
Driving	Unchanged
Reversing onto a difficult dock	Unchanged
Deciding conditions aren't safe	Unchanged

Human strengths that will matter most

1. **Vehicle handling in difficult conditions** — the dock nobody can reverse onto, the weather, the load that shifts. 2. **Judging safety** — the decision not to proceed, which carries real accountability. 3. **Customer-site problem-solving** — the delivery where access isn't what the system said. 4. **Specialist handling** — hazmat, oversize, livestock, abnormal loads. The specialisms automation reaches last and pays best. 5. **Local knowledge** — which matters more in multi-stop work than any planning system yet captures.

And underneath all five, two traits the score can't grade: curiosity and adaptability. The vehicles and the systems around them are changing — telematics, driver-assist, and above all electrification. The people who compound are the ones who stay genuinely curious and re-skill fastest: into specialist endorsements, into the high-voltage powertrain work that barely existed at scale three years ago, rather than defending the one segment they started in. In an industry where the desk roles are automating and the vehicle roles are shifting under electrification, that adaptability is itself a form of protection — and it is increasingly what the better employers hire for: the trajectory toward specialist and technical work, not the single ticket someone holds today.

Other things to consider about this job

What's genuinely good about it

The demand picture is strong and demographically driven.

- BLS projects roughly **237,600 annual openings** for heavy and tractor-trailer drivers through 2034
- The ATA projects a shortfall of around **82,000 drivers in 2026**, rising toward 160,000 by the end of the decade
- The average driver is around **46 years old**, and the ATA estimates the industry needs **1.2 million new drivers over the next decade** — roughly 120,000 a year — simply to replace retirements

Pay has risen materially and the increases have stuck. Driver wages are up around 18% since 2020, with growth of 15.5% in 2022, 7.6% in 2023 and 2.4% in 2024. Union drivers in the less-than-truckload segment average around **\$85,000** against **\$65,000** non-union, and a mid-level over-the-road driver on \$70,000 base costs a carrier \$87,500–\$94,500 fully loaded.

And the entry route is unusually accessible: short licensed training, frequently employer-funded, no degree, and no tuition debt.

What drivers report as rewarding: autonomy during the working day; being paid to be competent rather than to be present; a clear route to owner-operator status; and — in local and specialist work — genuine variety.

What's genuinely hard about it

Turnover is extraordinary, and it tells you something. Large truckload carriers report annual driver turnover of **90–95%**, and around **35% of newly hired drivers quit within their first 90 days.**

That figure is the honest counterweight to the shortage narrative, and it deserves stating plainly. University of Pennsylvania economist Steve Viscelli argues that the real problem is **retention rather than an absolute shortage of people willing to drive** — that an industry losing nine in ten of its workforce annually has a job-quality problem rather than a recruitment problem.

We think both readings are true and they point to the same practical advice: the demand is real, and the segment you choose determines whether the job is sustainable. Long-haul over-the-

road work has the worst conditions and the highest turnover. Local, LTL and specialist work has better pay growth and people stay.

The lifestyle cost in long-haul is genuine — multiple nights a week away from home, and that is the dominant reason people leave.

Injury rates run above average, at around 2.8 per 100 workers.

And the policy environment is unusually volatile right now, with visa restrictions and English-proficiency requirements affecting driver supply — a reminder that this sector is shaped by regulation as much as by technology.

Who this work suits — and who it doesn't

Transport work tends to suit people who:

- **Want autonomy during the day** — most of this work is unsupervised, which some people find liberating and others isolating
- **Are practically competent** and can solve a problem without calling someone
- **Are comfortable with responsibility** for an expensive vehicle and a serious safety duty
- **Prefer being paid for competence** rather than for presence
- **Are interested in eventually owning** — owner-operator is a normal destination

It's harder for people who:

- Need **social contact** through the working day
- Want **predictable home life** in long-haul work specifically
- Find **sitting for long periods** physically difficult
- Assume **all driving jobs are equivalent**. They are not, and the differences are large.

What else is moving this market

The demographic case is the durable one — an ageing workforce retiring faster than replacements arrive, and it holds regardless of the freight cycle.

Freight is cyclical, and the industry lost 122,000 positions from its 2022 peak before the current recovery.

And there is a growth area worth naming. Zero-emission vehicle registrations rose around 40% year-on-year, creating demand for high-voltage powertrain technicians — a role that

barely existed at scale three years ago. Vehicle maintenance is becoming more technical, not less, and those technicians are short everywhere.

The AI-native advantage — what to actually do about it

The situation: the technology arriving in transport is mostly arriving *around* the driver rather than replacing them — and the roles coordinating that technology are the exposed ones.

WHAT AN AI-NATIVE TRANSPORT PROFESSIONAL LOOKS LIKE

- 1. Choose the protected segment deliberately.** Local, specialist, emergency and multi-stop work is where autonomy reaches last and where pay growth has been strongest.
- 2. Specialize.** Hazmat, oversize, tanker, livestock, abnormal loads. Higher pay, and the categories autonomy addresses least.
- 3. Move toward the vehicle, not the desk.** Dispatch, planning and brokerage look like promotions. They score 6.8 and 7.3 against driving's 2.8 to 4.5.
- 4. Consider the technical route.** High-voltage powertrain and electric vehicle maintenance is a genuine shortage area, and it is physical, diagnostic work in unpredictable conditions — which is exactly the protected profile.
- 5. Plan for ownership.** Owner-operator status is the realistic route to the top of this industry's income range.

CONCRETE PREPARATION

Before committing: ride along with someone doing the specific job. Long-haul, local delivery and specialist work are genuinely different lives, and the difference is not obvious from outside.

Getting in: compare carriers on turnover rate and home time, not on sign-on bonus. With 90–95% turnover industry-wide, choosing the employer matters more than choosing the industry.

And ask about training funding — many carriers fund CDL training, which removes the entry cost entirely.

The routes in — and the versions that hold up

ROUTE	ASSESSMENT
CDL training → local or LTL driving	Short, often employer-funded, and the better-protected segment.
Specialist endorsements — hazmat, tanker, oversize	Higher pay, and the work autonomy reaches last.
Emergency & specialist vehicle operation	Best score in this guide at 2.8.
Commercial pilot training	Expensive and long, but protected by regulatory architecture.
Heavy vehicle / EV technician	Physical, diagnostic, chronically short. An under-considered route.
Logistics degree → planning and dispatch	The exposed end. Points at the 6.8–7.3 tracks.

Where this can lead later

Driving across segments, specialist and hazmat work, owner-operator and fleet ownership, driver training and assessment, safety and compliance management, vehicle technology and maintenance, and operations leadership.

One caution that runs against intuition. The conventional progression — driver to dispatcher to planner to operations manager — moves toward the exposed end of this industry. Every step off the vehicle raises exposure. That doesn't make it wrong, but it should be a deliberate choice rather than an assumed promotion.

What to look for in an employer or program

1. **Turnover rate.** The single most informative number and the one least advertised. Ask directly.
2. **Home time,** stated concretely rather than as a policy.
3. **Segment.** Long-haul, LTL, local, specialist — these are different jobs with different scores.

4. **Training funding**, and whether it carries a contractual tie-in.

5. **Equipment and maintenance standards**, which say a great deal about how a carrier treats its drivers.

Questions to ask a carrier or training provider

- What's your annual driver turnover rate?
- What proportion of new drivers are still here at twelve months?
- How many nights a week away from home, realistically?
- Do you fund CDL training, and what's the tie-in period?
- What endorsements do you support, and do you pay for them?
- What's the average age of your fleet?

Red flags: evasiveness about turnover; home time described vaguely; heavy emphasis on sign-on bonus over pay structure; long tie-in periods on funded training.

The career-hunting checklist

Choosing well is step one. This is what turns a good choice into a good outcome once you're in — worth reading now, because the best time to pick a segment and a carrier is knowing exactly what you'll need to do with it. The first list is for training and the early years; the second is for the job hunt that follows.

During training and the early years — build toward the protected segment

- **Choose the segment on purpose, not by whoever's hiring hardest.** Local delivery (3.4), specialist and emergency vehicle work (2.8) sit well below long-haul (4.5) and are where the pay growth and the retention are. The generic over-the-road seat is the exposed, high-turnover lane.
- **Stack specialist endorsements while entry is cheap.** Hazmat, tanker, oversize, livestock — higher pay, and the categories autonomy addresses last. These are the fastest way to move down the score table.

- **Treat the vehicle as the protected side of the industry.** Every step off it toward dispatch, planning or brokerage (6.8–7.3) raises exposure. If you want to progress, progress into specialist driving, training, safety, or ownership — not the desk.
- **Consider the heavy-vehicle and EV technician route seriously.** Zero-emission registrations rose around 40% year-on-year, and high-voltage powertrain technicians are short everywhere. It is physical, diagnostic work in unpredictable conditions — exactly the protected profile.
- **Feed curiosity and adaptability on purpose.** The vehicles and systems are shifting under electrification and telematics; the person who re-skills fastest into specialist and technical work compounds.
- **Keep the frame wide.** Vet the whole map — local, LTL, specialist, emergency, the technician route, ownership — before assuming long-haul is what "trucking" means. The most-advertised lane is almost always the highest-turnover one.

During the job hunt — choose the segment and the carrier, don't just take the seat

- **Target the protected segments on purpose.** Local, LTL, specialist and emergency work — not "long-haul, open to anything," which points at the highest-turnover, most-exposed driving lane.
- **Compare carriers on turnover and home time, not sign-on bonus.** With 90–95% turnover industry-wide and 35% of new drivers gone within 90 days, the carrier matters more than the industry. A large sign-on bonus is often the tell of a churn-and-burn operation.
- **Don't grab a desk role just because it's on offer and looks like a promotion.** A role being available doesn't make it safe. Dispatch, planning and brokerage score 6.8–7.3, and an operation still building out a large planning or dispatch team may simply not have adopted the optimization tooling yet — availability can be a lagging indicator, not a green light, and that's exactly the function offloaded at 30–50% overhead once they catch up.
- **Prefer the specialist seat, and the segment where people stay.** A hazmat, tanker or oversize role — or local and LTL work with real home time — puts you on firmer ground than the over-the-road seat that's easiest to get.
- **Ask the employer where the technology is heading in the role.** Whether they're investing in the driver — specialist work, better equipment, the electric fleet — or running the highest-turnover lane as cheaply as possible tells you if the seat grows or grinds you out.

- **Weigh the trajectory over the sign-on bonus.** A slightly lower-paid seat that builds endorsements, home time and a route to specialist or technical work beats a high-bonus long-haul seat in the lane autonomy targets first.

Go deeper — further reading

- **[Heavy and Tractor-trailer Truck Drivers — Occupational Outlook Handbook](#)** — US Bureau of Labor Statistics. The source of the 237,600 annual openings figure, plus wage data and entry requirements. Free, and the least commercially interested source available.
- **[American Trucking Associations — driver shortage research](#)** — the industry's own workforce analysis, including the shortfall projections and the 1.2 million replacement-hiring estimate. Read it knowing the ATA represents carriers.
- **[Truck driver shortage 2026: the real data](#)** — the clearest account of where autonomous trucking actually stands, including the fewer-than-50 driver-out trucks figure and the structural limits of the transfer-hub model.

The counterargument — read this one:

There are two, and they cut in opposite directions.

Our score is too low: autonomy is a trajectory, not a state. Fifty trucks today can be fifty thousand within a career, and long-haul highway driving is the single most automatable physical job in this index. [Entering long-haul work today is making a forty-year bet against a technology that is improving.](#)

Our score is too high: the shortage is severe, the workforce is ageing, 60% of the work is structurally out of autonomy's reach, and the industry cannot recruit fast enough to replace retirements.

And a third reading that questions the framing entirely. Economist Steve Viscelli argues the driver "shortage" is really a retention crisis — [90–95% annual turnover at large carriers is not what a labor shortage looks like; it is what a bad job looks like.](#) On that reading, the sector could staff itself tomorrow by improving conditions, and the shortage narrative serves carriers who prefer recruitment to retention.

Where we land: we think the retention critique is largely correct and doesn't change the advice. Demand for drivers is real either way, and the practical conclusion is the same —

choose the segment carefully, because the segments with the worst turnover are also the ones autonomy targets first.

Bottom line

The most confidently repeated prediction about any career in this index has not happened, and the opposite one is happening quietly — which is exactly why which version of transport you choose carries so much weight.

Nine years after autonomous trucking became the standard example of AI displacement, there are fewer than 50 driver-out trucks operating against roughly 4 million Class 8 trucks nationally — while dispatch, route planning and freight brokerage have been substantially automated, and can reportedly be offloaded at 30–50% overhead reduction.

The person behind the wheel is better protected than the person planning their route.

The protection is a timeline, not a wall, and honesty requires saying so. Long-haul highway driving is the most exposed driving category and someone entering it today should expect change within their career. Local, specialist, emergency and multi-stop work is on much firmer ground — and it is where the pay growth has been.

The honest problems have nothing to do with robots. Turnover of 90–95% at large carriers, with 35% of new drivers gone within 90 days, describes a job-quality problem rather than a technology one.

So the useful advice is: choose the segment deliberately rather than taking whichever job appears; specialize into hazmat, tanker or oversize where the pay is better and autonomy reaches last; compare carriers on turnover and home time rather than sign-on bonus; consider the heavy-vehicle and EV technician route, which is short of people and physically protected; and understand that moving to dispatch or planning is a step toward the exposed end, not away from it.

And one thing worth saying to anyone who has been told this career is finished. It isn't — but the reason it isn't has nothing to do with robots being bad at driving. It's that most of this work happens at loading docks, on customer sites and in city streets, and that part was never what anyone was automating.

Questions worth sitting with

Whether you're the one choosing or a parent thinking it through together, these are the conversations that lead to a better decision. Read "you" as whoever the choice is about.

About the work itself

1. What made transport work appealing? Push past "I like driving" to something specific. 2. **Do you understand that long-haul, local and specialist work are completely different lives?** Most people don't until they're in one. 3. How would you find being alone for most of the working day — liberating or isolating? 4. Are you comfortable being responsible for an expensive vehicle and a real safety duty?

Reacting to the findings

5. Fewer than 50 driver-out trucks against 4 million Class 8 trucks. Was that surprising? 6. **Logistics planning scores 7.3 and driving 2.8–4.5.** Does it change anything to know the office is automating faster than the cab? 7. The guide argues pilots are protected by regulation rather than manual skill. Does that idea apply anywhere else you're considering?

The harder conversation

8. **Turnover at large carriers runs 90–95%, and 35% of new drivers leave within 90 days.** What does that tell you about choosing an employer? 9. Long-haul means multiple nights a week away. Have you pictured that concretely? 10. The guide says the protection is a timeline rather than a wall. How do you feel about a long career in a sector where the technology is improving?

Testing it against reality

11. **The most valuable single action:** ride along for a full day in the specific segment you're considering. Long-haul and local delivery are different jobs and the difference isn't visible from outside. 12. Find a driver ten years in and ask what their body feels like, and whether they'd do it again.

Widening the frame

13. Have you considered **specialist endorsements** — hazmat, tanker, oversize? Better pay, and the work autonomy reaches last. 14. What about **heavy vehicle or EV technician** work? Physical, diagnostic, chronically short of people. 15. Have you looked at **emergency and specialist vehicle operation**? Best score in this guide at 2.8.

This is analysis and informed judgment about a fast-moving situation. For transport specifically, we've flagged that autonomy is a genuine trajectory on an uncertain timeline, that the sector's turnover figures suggest a job-quality problem our framework cannot see, and that the roles coordinating transport are considerably more exposed than the roles performing it.

Scores for 2023 and 2025 are retrospective reconstructions using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Transport

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned} \text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment}) \end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

EMERGENCY & SPECIALIST VEHICLE OPERATION — 2.8 (VERY LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.8	5.2	5.7	1.99
Entry-path erosion	15%	1.5	1.8	2.0	0.3
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	8.5	8.5	8.5	0.22
Regulatory / licensing	10%	9.0	9.0	9.0	0.1
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	2.8 → 2.8

LOCAL DELIVERY & MULTI-STOP — 3.4 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.6	5.1	5.6	1.96
Entry-path erosion	15%	1.8	2.0	2.2	0.33
Physical / embodied	15%	9.3	9.3	9.3	0.1
Human trust / accountability	15%	7.0	7.0	7.0	0.45
Regulatory / licensing	10%	6.5	6.5	6.5	0.35
Judgment under novelty	10%	8.0	8.0	8.0	0.2
				Total	3.4 → 3.4

AIRLINE PILOT — 3.6 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.9	6.6	7.1	2.48
Entry-path erosion	15%	2.5	2.8	3.0	0.45
Physical / embodied	15%	8.0	8.0	8.0	0.3
Human trust / accountability	15%	8.5	8.5	8.5	0.22
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	3.61 → 3.6

RAIL OPERATION — 4.2 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.2	6.9	7.7	2.69
Entry-path erosion	15%	2.2	2.5	2.8	0.42
Physical / embodied	15%	8.5	8.5	8.5	0.22
Human trust / accountability	15%	7.5	7.5	7.5	0.38
Regulatory / licensing	10%	8.5	8.5	8.5	0.15
Judgment under novelty	10%	6.5	6.5	6.5	0.35
				Total	4.21 → 4.2

LONG-HAUL TRUCKING — 4.5 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.6	6.7	7.3	2.55
Entry-path erosion	15%	2.5	2.8	3.2	0.48
Physical / embodied	15%	9.0	9.0	9.0	0.15
Human trust / accountability	15%	6.0	6.0	6.0	0.6
Regulatory / licensing	10%	6.5	6.5	6.5	0.35
Judgment under novelty	10%	6.5	6.5	6.5	0.35
				Total	4.48 → 4.5

FREIGHT BROKERAGE & DISPATCH — 6.8 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.5	7.0	7.9	2.77
Entry-path erosion	15%	4.5	5.0	5.5	0.82
Physical / embodied	15%	1.5	1.5	1.5	1.27
Human trust / accountability	15%	5.5	5.5	5.5	0.67
Regulatory / licensing	10%	2.5	2.5	2.5	0.75
Judgment under novelty	10%	5.0	5.0	5.0	0.5
				Total	6.79 → 6.8

LOGISTICS PLANNING & SCHEDULING — 7.3 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.7	7.5	8.3	2.91
Entry-path erosion	15%	5.0	5.6	6.2	0.93
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	5.0	5.0	5.0	0.75
Regulatory / licensing	10%	2.0	2.0	2.0	0.8
Judgment under novelty	10%	4.5	4.5	4.5	0.55
				Total	7.28 → 7.3

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Emergency & specialist vehicle operation

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	4.8	1.5	0.5	2.4
2025	5.2	1.8	0.5	2.6
Fall 2026	5.7	2.0	0.5	2.8

Movement decomposition: automatability contributed **+0.32** and entry-path erosion **+0.07**, for a total of **+0.4** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.